



MECHANICS OF FLIGHT I 25/26-S2

Lab - Group 28

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Introduction

This document reports on the first laboratory session on Flight Mechanics. The lab itself centers around the dynamics of a system that models the behavior of a rudder in an aircraft, with an application of spring forces. The objective is to understand how changing the position of this stabilizing element with respect to the center of mass of the aircraft helps decide the configuration on real aircraft.

1 Analytical Calculations

Question	Answer
1	$f = \begin{pmatrix} \dot{\psi} \\ \Omega^2(\delta \cos \psi - 1)\psi \end{pmatrix}$
2	$u_0 = [0 \ 0]$ $u_+ = [\arccos(1/\delta) \ 0]$, u_+ exists if $x_p > 0.05787$ $u_- = [-\arccos(1/\delta) \ 0]$, u_- exists if $x_p > 0.05787$
3	$\left. \frac{\partial M_{OE}}{\partial \psi} \right _{u_0} = \frac{1}{2} \rho S U_\infty^2 a_w - k = 1.728 x_p - 0.1$
4	if $\left. \frac{\partial M_{OE}}{\partial \psi} \right _{u_0} < 0$ then u_0 is stable if $\left. \frac{\partial M_{OE}}{\partial \psi} \right _{u_0} > 0$ then u_0 is unstable
5	$x_p^* = \frac{2k}{\rho S U_\infty^2 a_w}$ (from $\frac{\partial M_{OE}}{\partial \psi} = 0$)
6	$\vec{J} = \begin{pmatrix} 0 & 1 \\ \Omega^2(\delta - 1) & 0 \end{pmatrix}$
7	$\lambda_1 = \Omega\sqrt{\delta - 1}$ $\lambda_2 = -\Omega\sqrt{\delta - 1}$
8	u_0 is stable if $\delta < 1$
9	$x_p^* = 0.05787 \text{ m}$

The computation of equilibrium positions can be done by declaring the second term of the matrix to be equal to 0. Apart from the trivial solution, 2 others exists only **if** $\delta > 1$, which can be particularized as $x_p > 0.05787$.

For the stability, we know that the positive derivative of the moment will result in instabil-

ity. The physical reasoning behind this is that if the airfoil is turned upward, and a moment is applied upward as well, the motion will continue until a stall or some other stopping event occurs. If the derivative is negative, it is interpreted as a restoring moment that pulls back toward the equilibrium position.

It is important to note that the discussion of stability in the second to last question, the Jacobian's eigenvalues were calculated at u_0 . These tell us that whenever $\delta > 1$ the eigenvalues are purely real, but when $\delta < 1$ then they are purely complex. In the first case, the trivial solution corresponds to a saddle point, which means that it is an unstable equilibrium point. In the second case, the same point is now stable and forms periodic orbits around it, as will be reflected in the extracted figures.

2 Figures and Trajectories

What follows in this section are the plots against time and the phase space diagrams for the x_p values: $x_p = 0.1$ ($x_p > x_p^*$) and $x_p = 0.04$ ($x_p < x_p^*$). As mentioned above, in figure 3 we observe a saddle point, and in figure 6 we observe a center, agreeing with our previous conclusions.

One can also see by relating both the $\psi - \dot{\psi}$ and $\psi - t$ diagrams that the smaller the orbits, the smaller the period

Lastly, it is relevant to mention that the random initial conditions vary within the range of $[-3, 3]$ to ensure the reproducibility of the results.

2.1 Case 1: $x_p = 0.1$ (Unstable u_0)

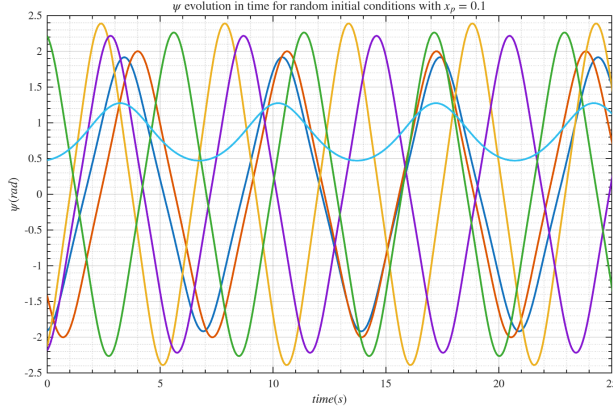


Figure 1: $\psi - t$ for $x_p = 0.1$.

2.2 Case 2: $x_p = 0.04$ (Stable u_0)

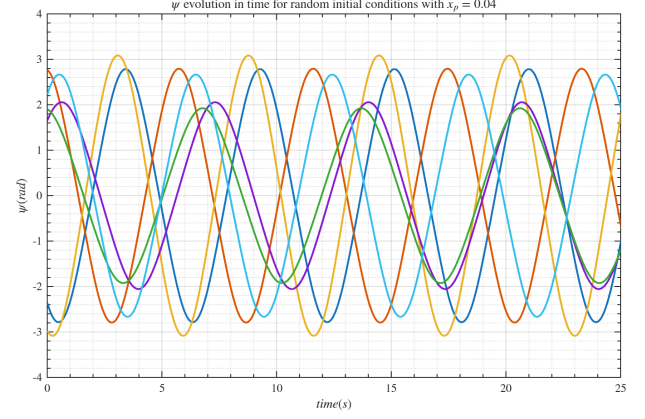


Figure 4: $\psi - t$ for $x_p = 0.04$.

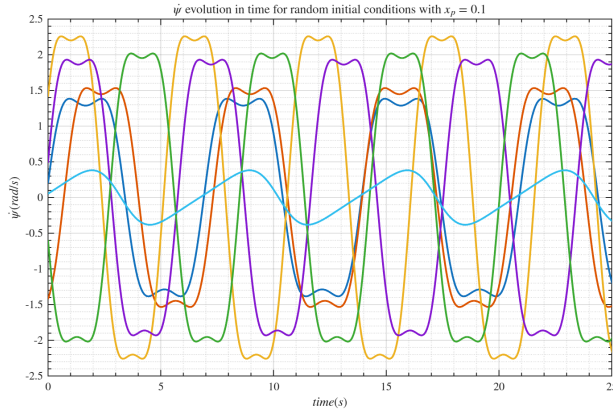


Figure 2: $\dot{\psi} - t$ for $x_p = 0.1$.

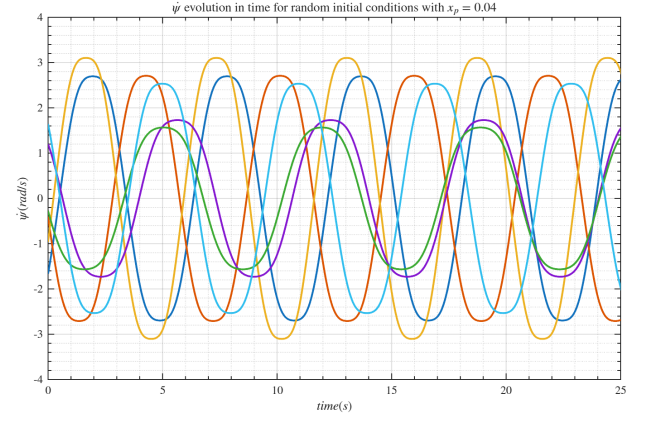


Figure 5: $\dot{\psi} - t$ for $x_p = 0.04$.

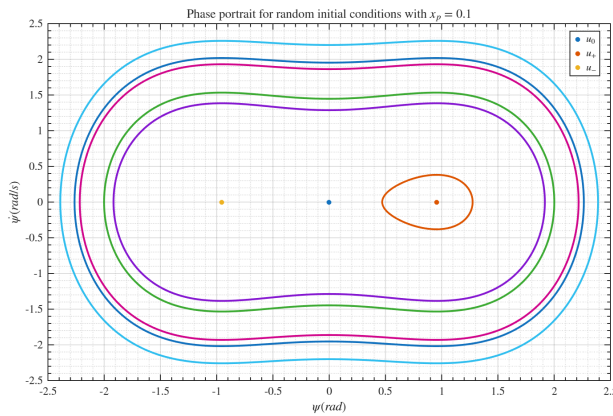


Figure 3: $\psi - \dot{\psi}$ for $x_p = 0.1$.

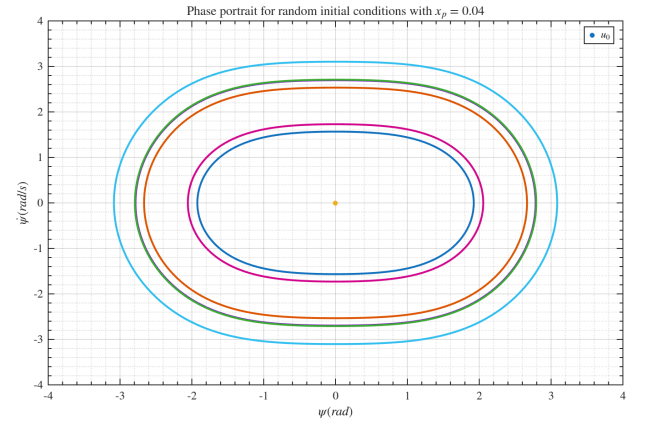


Figure 6: $\psi - \dot{\psi}$ for $x_p = 0.04$.

3 Discussion of the Results

For the case where $x_p < x_p^*$ ($x_p = 0.04$), the trivial solution is stable. Closed, full orbits are visible around it. Physically, this signifies that the torque generated by the airfoil is larger than the spring's at any given angle, leading to only one stable point remaining at the origin (our trivial solution).

On the other hand, for $x_p > x_p^*$ ($x_p = 0.1$), u_0 is unstable, although still an equilibrium point. The newly found equilibria are stable, and orbits are formed either fully around them, or only around each, depending on the magnitude of the initial conditions. For their physical meaning, they are the points where the net summation of both torques produced by the spring and lift is equal to 0.

4 Relation with common aircraft Configuration

It is very clear that almost every aircraft in flight today has a vertical stabilizer either on its tail, or at the end of its body, be it a delta-wing aircraft. This is, usually the vertical stabilizer will be found behind the center of mass of the vehicle.

Were the vertical stabilizer be placed at the front of the aircraft, even small perturbations would make the airplane turn dramatically from front to back due to the torque produced. Of course, if it is placed at the tail this does not occur. This is due to the fact that if the plane is now hit by a perturbation the torque caused by the tail will correct its direction.

This can also be found in a variety of different configurations, not only vertical or even upward, as with the General Atomics MQ-1 Predator, the

Beechcraft Bonanza, or many others.

5 Effect of Dissipation

Dissipation is introduced as a damping term in the dynamics, modeling either friction or aerodynamic damping in the system.

Now the phase diagrams show that orbits are no longer present. For $x_p < x_p^*$ ($x_p = 0.04$), every possible initial condition will collapse toward the origin, which is now an attractor, since the eigenvalues now have a negative real part and imaginary one. For $x_p > x_p^*$ ($x_p = 0.1$), trajectories will be repelled from the origin and converge in either of the other equilibria, but will never converge toward the saddle point in the origin.

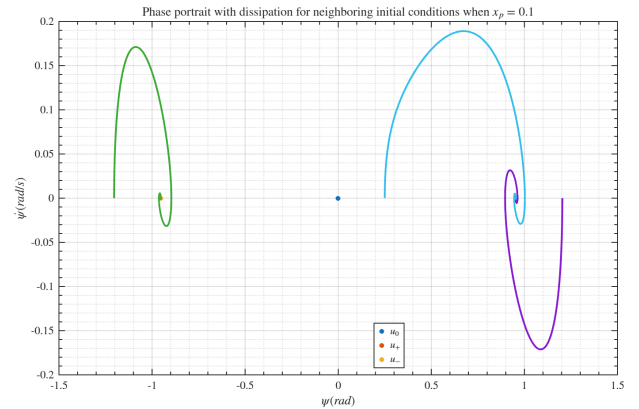


Figure 7: Phase space diagram with dissipation for $x_p = 0.1$.

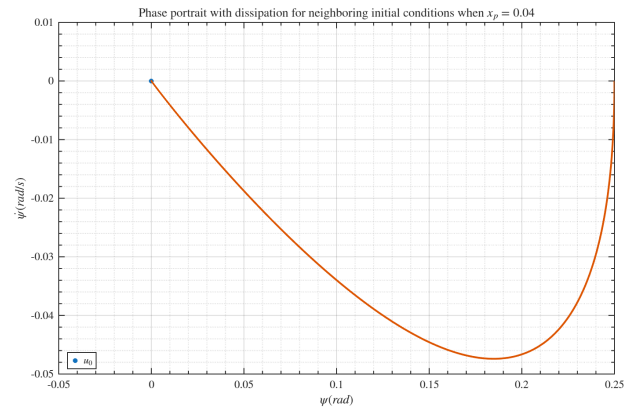


Figure 8: Phase space diagram with dissipation for $x_p = 0.04$.